

Detecting Informed Trading Before Unscheduled Geopolitical Announcements: Evidence from Cross-Asset Futures Volume

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April 2026 — Version 10 — Reproducible analysis notebook attached

Abstract

This paper develops and applies a cross-asset informed-flow detection framework to CME ES (S&P 500 E-mini) and CL (WTI crude oil) futures surrounding six unscheduled geopolitical announcements by the Trump administration during the Iran conflict of March–April 2026. Using 1-minute OHLCV bars sourced from Databento (GLBX.MDP3), the framework identifies bars where simultaneous volume anomalies in both instruments are directionally consistent with the content of the subsequent announcement. Applied to six cataloged events, the framework detects cross-asset spikes that are statistically unlikely under a random-timing null preceding two de-escalation announcements: E004 (CL $Z = 11.93$, return -31.5 bps, $T-16$ minutes) and E007 (ES $Z = 8.54$, CL $Z = 14.01$, return -244.7 bps on CL, $T-167$ minutes). The findings are robust across 320 parameter combinations (100% confirmation rate, 0% false positive rate on control events). A within-window Fisher combined test yields $p = 0.0035$. The analysis demonstrates that publicly available 1-minute exchange data can detect cross-asset informed flow before unscheduled geopolitical announcements without access to position-level or order-flow data. All code is reproducible with a free Databento API key.

1. Introduction

Between March and April 2026, the Trump administration issued a series of public announcements related to a US military conflict with Iran. These announcements caused sharp movements in equity and energy futures markets: oil prices fell as much as 15% on de-escalation posts and equities rose correspondingly. The scale and speed of these moves created an unusually high-value information environment for any trader with prior knowledge of announcement content. This

episode therefore provides a rare natural experiment in which the timing of information release is precisely observed ex post, but not known ex ante, making it well-suited for testing a scanning-based informed-flow detection methodology.

The analysis contributes to the literature in three ways. First, it demonstrates that publicly available 1-minute exchange data can detect cross-asset informed flow without access to position-level or order-flow data, extending the methodology of Bernile, Bhagwat, and Rau (2016) from scheduled macro announcements to unscheduled, high-stakes geopolitical events. Second, it introduces cross-asset directional confirmation as a filter for volume anomalies: a spike that is consistent across ES and CL in the direction predicted by announcement content provides stronger evidence of informed positioning than a single-asset volume shock. Third, it establishes a reproducible, parameter-robust detection framework that can be applied to any discrete unscheduled event for which public 1-minute futures data exist, requiring only a free exchange data API.

These events also attracted external regulatory and journalistic attention, providing an additional point of contact between the detection framework and independently documented market activity. The present analysis relies solely on publicly available exchange data and does not speak to the identity or scale of any positions involved.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 details the methodology. Section 5 reports results. Section 6 presents the statistical assessment. Section 7 discusses limitations. Section 8 concludes.

2. Related literature

2.1 Informed trading before announcements

The theoretical foundation for this paper is Kyle (1985), who shows that an informed trader with private information about an impending price-relevant event will gradually accumulate a position in the periods before that event becomes public, generating abnormal volume as the price moves toward its post-announcement level. The volume spike is a necessary but not sufficient condition for informed trading; the direction of the price move must also be consistent with the information content of the event.

A large empirical literature documents informed trading ahead of scheduled corporate and macroeconomic announcements. Kecskes, Michaely, and Womack (2010) find abnormal short-selling activity in the days preceding analyst downgrades, consistent with tipping. Augustin, Brenner, and Subrahmanyam (2019) document anomalous option volumes preceding roughly 25% of all takeover announcements, concentrated in short-dated out-of-the-money calls. Bernile, Bhagwat, and Rau (2016) use high-frequency E-mini S&P 500 futures data to detect informed trading during FOMC embargo periods, finding that abnormal order imbalances are in the direction of subsequent policy surprises. That paper is the closest structural antecedent to the present

analysis: it applies a volume-based detection methodology to E-mini futures around an announcement event. The present paper extends this approach in two respects. The announcements studied here are unscheduled, so the standard event-study assumption of a known announcement time is replaced by a scanning approach. The cross-asset confirmation filter (ES and CL jointly) is also new; Bernile et al. rely on a single-instrument order imbalance measure.

The distinction between scheduled and unscheduled events matters for statistical design. Scheduled announcements permit precise pre/post windows and comparison across many events. Unscheduled geopolitical posts require a scanner that identifies anomalous activity within a broad window, then tests whether identified spikes are directionally consistent. The within-window Fisher test used here (Section 6.1) is adapted from the permutation approach recommended by Phipson and Smyth (2010) for exactly this setting.

2.2 Informed trading venue choice

When informed traders possess private information about a macro event, they must choose which instrument to trade. Easley, O'Hara, and Srinivas (1998) show theoretically that informed traders migrate to options when the leverage advantage outweighs the higher transaction costs; this prediction has substantial empirical support for corporate events. For macro and geopolitical events, however, index futures dominate as the informed-trading venue. The E-mini S&P 500 contract (ES) is the most liquid single instrument for expressing a view on broad market direction. WTI crude oil futures (CL) are the primary instrument for expressing a view on oil supply and geopolitical risk. The joint use of ES and CL to express a de-escalation view (long equities, short oil) is the structural pattern this paper detects.

Cao, Chen, and Griffin (2005) document option-market informed trading before September 11, 2001, finding put buying in airline and insurance stocks in the days preceding the attacks. Their methodology—comparing pre-event option activity to a normal baseline—is an early application of the event-study approach to unscheduled geopolitical events. The present paper uses a volume Z-score normalized against a 60-bar rolling baseline, which is methodologically similar but applied to 1-minute futures bars rather than daily option volumes.

2.3 Cross-asset information transmission

The cross-asset nature of the detected signal—simultaneous ES and CL anomalies in opposite directions—connects this paper to the literature on cross-asset information transmission. Hasbrouck (1995) develops the information-share framework for measuring price discovery across related markets. Applied to ES and CL, the framework would ask which instrument leads in incorporating the geopolitical information. This paper does not compute formal information shares; instead, it uses co-occurrence of anomalous volume in both instruments as a filter for informed positioning rather than noise.

The directional filter is motivated by a simple observation: a volume spike in ES alone could reflect algorithmic momentum or broad institutional rebalancing. A spike in CL alone could reflect an oil-specific news release. A simultaneous spike in both instruments, directionally consistent with the same subsequent announcement, is far harder to attribute to coincidence. The sensitivity analysis (Section 6.3) confirms that this cross-asset requirement eliminates all false positives across 320 parameter combinations.

2.4 Geopolitical events and commodity futures

A growing literature documents the impact of geopolitical risk on commodity and equity markets at lower frequency. Caldara and Iacoviello (2022) construct a news-based geopolitical risk index and show it predicts crude oil price volatility and equity return distributions at monthly frequency. Smales (2021) finds that geopolitical risk shocks increase CL futures volatility and reduce liquidity. These papers operate at horizons where the information channel is diffuse: geopolitical news aggregates over days or weeks before prices adjust. The present paper examines a different phenomenon: a single, discrete, unscheduled announcement at a known time, with the hypothesis that a specific informed trader possessed advance knowledge of the content. This discrete event-study design is distinct from the geopolitical risk index literature and requires the microstructure methodology described below.

3. Data

3.1 Source

Data are sourced from Databento (GLBX.MDP3), the CME Group market data feed. Schema: ohlcv-1m (1-minute OHLCV bars). Instruments: ES (S&P 500 E-mini) and CL (WTI crude oil) front-month futures. The dataset spans 1,780 ES bars and 1,720 CL bars across targeted windows from February 6 to April 9, 2026.

3.2 Event catalog

Six events were cataloged from public reporting. E002 and E003 are Saturday posts when CME equity futures are closed and therefore undetectable by this methodology. E004 and E007 are the primary events of interest.

Table 1. Event catalog. Six Trump Truth Social posts related to the Iran conflict, March–April 2026. Saturday posts (E002, E003) are undetectable because CME equity futures are closed.

ID	Date / Time ET	Description	Type	Status
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E00 2	Mar 21 AM	02:43	Winding-down signal	DE- ESC	UNDETECTABLE (Saturday)
E00 3	Mar 21 AM	02:44	48-hr ultimatum	ESC	UNDETECTABLE (Saturday)
E00 4	Mar 23 AM	07:05	5-day pause	DE- ESC	CONFIRMED
E00 6	Mar 26 PM	~12:15	Cabinet — NO TURNING BACK	ESC	UNCONFIRMED (wrong direction)
E00 5	Mar 26 PM	04:11	10-day extension	DE- ESC	UNCONFIRMED (two-sided volume)
E00 7	Apr 7 PM	06:32	Two-week ceasefire	DE- ESC	CONFIRMED

4. Methodology

4.1 Volume Z-score

For each 1-minute bar, volume is normalized against a 60-bar rolling mean and standard deviation computed from prior bars only (a one-period lag prevents lookahead bias):

$$Z_t = (V_t - \mu_{t-1,60}) / \sigma_{t-1,60}$$

This formulation follows Kyle (1985): informed traders generate abnormal volume at the moment of private information arrival. A high Z-score is necessary but not sufficient; directional consistency with the subsequent announcement is required as a second filter, motivated by the cross-asset venue-choice argument in Section 2.2.

4.2 Cross-asset scanner

For each event window (180 minutes prior to announcement), the scanner identifies bars where both ES and CL Z-scores exceed thresholds within a co-occurrence window. Canonical parameters: ES threshold 1.5σ , CL threshold 1.2σ (lower, given CL's higher sensitivity to geopolitical news), co-occurrence window 15 minutes, maximum lead 180 minutes. Signals are deduplicated by retaining the highest ES Z per (event, CL spike time) pair.

4.3 Direction check

For each flagged signal, the price return at the spike bar is evaluated for directional consistency with the subsequent announcement content. De-escalation is expected to produce ES up and CL down (risk-on buying, falling supply-disruption premium). Escalation is expected to produce ES

down and CL up. Wrong-direction signals are classified as unconfirmed regardless of Z-score magnitude.

4.4 Baseline

Six Friday 3:59 PM ET market-on-close bars and three non-event Tuesday mid-session windows (April 1, 2, and 9) serve as the noise baseline. The March 20 MOC bar ($Z = 29.91$) is 1.87 standard deviations above the Friday mean, confirmed anomalous by one-sample t-test ($t = -4.57$, $p = 0.006$). Non-event Tuesday windows produce zero scanner signals.

4.5 Null hypothesis

Under the null hypothesis, volume spikes occur randomly within the pre-announcement window and are not systematically aligned with the directional implications of the subsequent announcement across both instruments. That is, the timing of any cross-asset volume co-occurrence is independent of announcement content, and the direction of the associated price moves is drawn from a distribution with equal probability of consistency and inconsistency. The statistical tests in Section 6 evaluate whether the detected spike bars deviate from this random-timing, random-direction benchmark.

5. Results

This section illustrates the behavior of the detection framework on the March–April 2026 event set.

5.1 E004 — March 23 five-day pause (confirmed)

At 6:49 AM ET (10:49 UTC), 16 minutes before the 7:05 AM ET announcement, a cross-asset spike produced CL $Z = 11.93$ with return -31.5 bps and ES $Z = 8.37$ with return $+0.4$ bps. The CL return at the spike bar is directionally consistent with a position that would benefit from the subsequent announcement-driven price move. The ES leg is present but small in return magnitude. The spike occurs within the pre-announcement window at a Z-score level that is anomalous relative to the Friday MOC baseline described in Section 4.4.

5.2 E007 — April 7 ceasefire (confirmed)

At 3:45 PM ET (19:45 UTC), 167 minutes before the 6:32 PM ET announcement, a cross-asset spike produced ES $Z = 8.54$ with return $+31.7$ bps and CL $Z = 14.01$ with return -244.7 bps ($-\$2.73$ per barrel in one minute). Both returns are directionally consistent with positions that would benefit from the subsequent de-escalation announcement. The lead time of 167 minutes is notable: the spike precedes the announcement during a period of publicly elevated geopolitical tension, which makes alternative explanations based on momentum or trend-following less directly aligned

with the observed pattern. The CL return of -244.7 bps in a single bar is the largest single-bar magnitude in the dataset and is consistent with concentrated directional trading activity at that timestamp.

5.3 Unconfirmed events

E006 (March 26 cabinet escalation): ES $Z = 6.69$ and CL $Z = 5.96$ are anomalous but both price directions are inconsistent with the escalation announcement that followed. The source timestamp is approximate, and the result is classified as noise. E005 (March 26 ten-day extension): ES $Z = 37.06$ is the highest Z-score in the dataset, but the return is -0.8 bps, near-zero despite massive volume. This pattern is consistent with two-sided institutional flow involving offsetting orders rather than directional pre-positioning.

These two events are also the ones that received external regulatory and journalistic attention, though the present analysis relies solely on publicly available exchange data and does not speak to the identity or scale of any positions involved.

Table 2. Summary of detected signals. Lead is measured from spike time to announcement time.

Event	Spike time ET	ES Z / return	CL Z / return	Lead (min)	Verdict
E004	06:49 AM Mar 23	8.37 / +0.4 bps	11.93 / -31.5 bps	16	CONFIRMED
E007	03:45 PM Apr 7	8.54 / +31.7 bps	14.01 / -244.7 bps	167	CONFIRMED
E006	11:46 AM Mar 26	6.69 / +9.9 bps	5.96 / -55.0 bps	89	UNCONFIRMED (wrong direction)
E005	03:59 PM Mar 26	37.06 / -0.8 bps	2.08 / +8.5 bps	12	UNCONFIRMED (two-sided)

6. Statistical assessment

6.1 Within-window timing test (Fisher's method)

For each event, the 180-minute pre-announcement window was examined to determine the fraction of bars with Z-scores at least as extreme as the spike bar and showing directional consistency. This fraction is the per-event p-value under the null of random timing.

Table 3. Per-event p-values for the within-window Fisher test. Base rates near 0.25 validate the binomial assumption.

Event	Consistent	Base rate	p-value	Bars in window
E004	Yes	0.27	0.0238	125
E006	No	0.33	0.0074	135
E005	No	0.20	0.0055	180
E007	Yes	0.32	0.0165	120

Combining the two confirmed events via Fisher’s method: $\chi^2 = 15.68$, $df = 4$, $p = 0.0035$. The joint probability that both spike bars appear this extreme in Z-score and show directional consistency by chance is 0.35%.

6.2 Binomial test

Treating direction consistency as a Bernoulli trial with conservative null $p = 0.25$, observing two consistent signals out of four detectable events yields one-sided $p = 0.26$. This does not reject the null in isolation. The minimum achievable p with $N = 4$ is 0.0039. The binomial test is reported for completeness; the Fisher within-window test is the primary statistical evidence.

6.3 Sensitivity analysis (320 parameter combinations)

The scanner was run across a full grid: rolling window $\in \{30, 45, 60, 90\}$ bars; co-occurrence window $\in \{5, 10, 15, 20, 30\}$ minutes; ES threshold $\in \{1.0, 1.5, 2.0, 2.5\}$; CL threshold $\in \{0.8, 1.0, 1.2, 1.5\}$. Results are uniform: E004 confirmed in 320/320 combinations; E007 confirmed in 320/320 combinations; both confirmed simultaneously in 320/320 combinations; false positives on E005 and E006 in 0/320 combinations. These results indicate that the findings are not sensitive to threshold choice. This parameter stability reduces concerns that the detected spikes arise from threshold tuning or researcher degrees of freedom.

7. Limitations

Four limitations bound the interpretation of the results.

Small sample. $N = 4$ detectable events. The binomial test alone cannot reject the null hypothesis of uninformed trading. The Fisher within-window test and the 320-combination sensitivity sweep are the primary statistical evidence, but independent replication on a larger event set is required before

strong causal claims are warranted. The purpose of this paper is therefore methodological demonstration rather than definitive statistical inference.

Data resolution. The 1-minute OHLCV bars used here aggregate all trades within each minute. Sub-second tick data would allow more precise lead-time estimation, order-flow decomposition (buy-initiated versus sell-initiated volume), and computation of formal information-share measures (Hasbrouck, 1995). The 1-minute bars are consistent with informed trading but cannot distinguish it from other sources of anomalous volume at fine timescales.

No trader identity. This analysis identifies anomalous market-wide behavior; it does not identify who placed the trades. Volume spikes can arise from algorithmic momentum, correlated institutional rebalancing, or reactions to other contemporaneous news sources. The cross-asset directional confirmation filter and the 100% parameter stability substantially reduce but do not eliminate these alternative explanations.

Generalizability. The Iran war announcements studied here represent an extreme information environment: large, discrete, binary events posted on a public social media platform by a sitting head of state. The informed-flow detection methodology may not generalize to lower-stakes or more diffuse geopolitical events where the information signal-to-noise ratio is lower.

8. Conclusion

This paper develops a cross-asset informed-flow detection framework for unscheduled geopolitical announcements and applies it to six events from the Iran conflict of March–April 2026. The framework detects volume anomalies that are unlikely under the random-timing null described in Section 4.5 preceding two de-escalation announcements: E004 (CL $Z = 11.93$, return -31.5 bps, 16-minute lead) and E007 (ES $Z = 8.54$, CL $Z = 14.01$, return -244.7 bps on CL, 167-minute lead). Both results hold across 320 parameter combinations with zero false positives, and the Fisher combined test yields $p = 0.0035$.

The observed patterns are consistent with the Kyle (1985) model of informed trading: both the volume anomalies and the directional structure of the price moves at the spike bars are consistent with positions that would benefit from the subsequent announcement content. The framework operates entirely on publicly available CME exchange data, requiring no proprietary order-flow records.

Three extensions would strengthen the analysis. Tick-level data would allow order-flow decomposition and formal information-share estimation. Expanding the event catalog to prior Trump geopolitical announcements (2017–2019) would provide out-of-sample tests of the detection methodology. Applying the cross-asset scanner to other high-stakes unscheduled events

(central bank interventions, surprise regulatory announcements) would assess how broadly the approach generalizes beyond the Iran conflict setting.

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Appendix: Reproducibility

All results are produced by the attached notebook (`informed_flow_v9.ipynb`). To reproduce: obtain a free Databento API key at databento.com, replace the key in cell 3, and run all cells. Total runtime is approximately 5–10 minutes. The notebook is self-contained; every number in this paper can be independently verified from the CME exchange record.

Note: rotate the Databento API key in cell 3 before sharing the notebook publicly.